

73-103 · CMU-Q

Principles of Macroeconomics

Fall 2026

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Business Administration

Meeting 1

Introduction and Motivation

Monday, 24 August

Outline

1

Introductions and Initial Background

2

Why should a business student study macroeconomics?

3

What does Macroeconomics study?

4

Structure, expectations, and AI in this class

BLOCK 1

Introductions and Initial Background

Connecting your interests to macroeconomics

01

Introductions

Share your name and background.

02

Expectations

What do you hope to gain from this course by the end of the semester?

03

Current events

Which business or economic problem has caught your interest over the last year? Why?

So far ... micro tools you already have

Supply and Demand

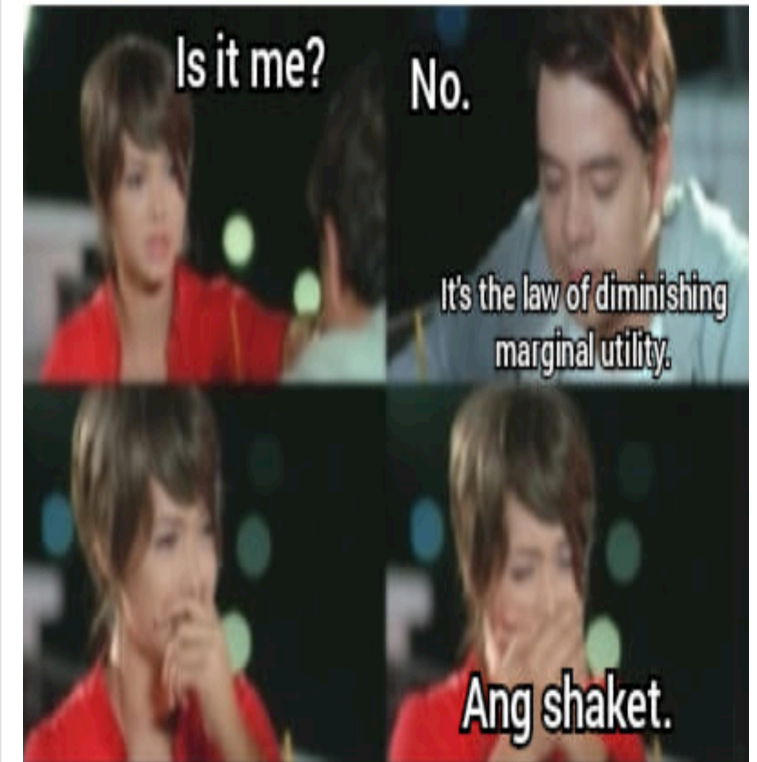


Opportunity Cost



Diminishing marginal utility

ECONOMISTS DURING BREAK-UPS



Macroeconomic problems

What do you think are the differences between microeconomic and macroeconomic problems?

THE WALL STREET JOURNAL.

ECONOMY | THE OUTLOOK

After Pulling Inflation Down, Gasoline and Food Threaten to Nudge It Up

Federal Reserve will worry if higher headline inflation feeds through to wage and price-setting

By [Gwynn Guilford](#) [Follow](#) and [Will Parker](#) [Follow](#)

Aug. 13, 2023 at 7:00 am ET

Gift unlocked article

Listen (6 min)



Inflation — gasoline and food

BUSINESS

UK economy faces reckoning over £2.6tn public debt pile

BLOOMBERG/LONDON

LAST EDITED AUGUST 19, 2023 | 11:30 PM



Skyscrapers and buildings on the City of London skyline. The UK's public debt load has soared by more than 40% to almost £2.6tn since the pandemic struck, leaving the country owing more than its entire annual economic output for the first time since 1961.

From the financial crisis to Russia's invasion of Ukraine, Britain has borrowed and spent its way out of

UK public debt · £2.6tn

BUSINESS

Global trade set to recover in coming months, says QNB

DOHA

LAST EDITED AUGUST 19, 2023 | 08:02 PM



Year	Global Trade Volume (index, 2010=100, seasonally adjusted)
2020	104.39
2021	130.21
2022	136.56
2023	131.78

Source: Haver, Netherlands BEPA, QNB Economics

Global trade is set to recover in the coming months, supported by foreign exchange movements, bottomed out international manufacturing cycle and improved Dow Jones transportation average, according to QNB. "In our view, leading indicators suggest that global

Global trade — QNB

Distinguishing micro from macro problems

Microeconomics

1

Focuses on individual agents: consumers, workers, and households.

2

Analyzes specific firms and how they make production and pricing decisions.

3

Studies individual markets (e.g., the market for oil or labor).

Macroeconomics

1

Analyzes aggregate variables: the total output of an entire nation.

2

Monitors economy-wide phenomena like inflation and unemployment.

3

Evaluates national policies and their impact on long-term growth.

Macro or micro?

U.S. Politics Economy **tech** Finance Opinion Arts & Culture Lifestyle R

TECH

Chip Inventories Swell as Consumers Buy Fewer Gadgets

Semiconductor companies slash production plans amid weak demand

By [Asa Fitch](#) [Follow](#)

Dec. 27, 2022 at 5:30 am ET

➦

🔖

AA

📺 Gift unlocked article

🔊 Listen

(6 min)

⋮



Chip inventories swell

ECONOMY | U.S. ECONOMY

Inflation Eased to 3% in June, Slowest Pace in More Than Two Years

Fed officials are still likely to raise interest rates when they meet later this month

By [Gwynn Guilford](#) [Follow](#) and [Nick Timiraos](#) [Follow](#)

Updated July 12, 2023 at 4:09 pm ET

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AA

💬 1517

📺 Gift unlocked article

🔊 Listen

(7 min)

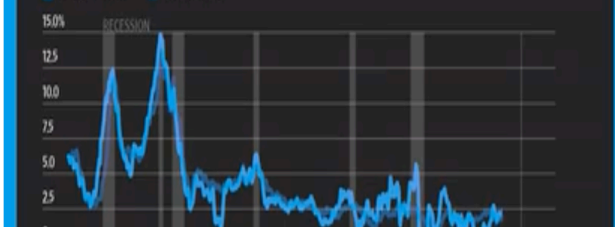
⋮

🔊 TAP FOR SOUND

WSJ

CONSUMER-PRICE INDEX, CHANGE FROM A YEAR EARLIER

■ Overall (+3.0%) ■ Core (+4.8%)



Inflation eased to 3%

THE WALL STREET JOURNAL

Why Your Steak Is Getting Pricier

Ranchers are shrinking cattle herds because of drought and high costs, cutting down the nation's supply of beef. That threatens to push prices for steaks and burgers to records.



WSJ

ISTO

Why your steak is getting pricier

73-103 · Intro · Mon 24 Aug

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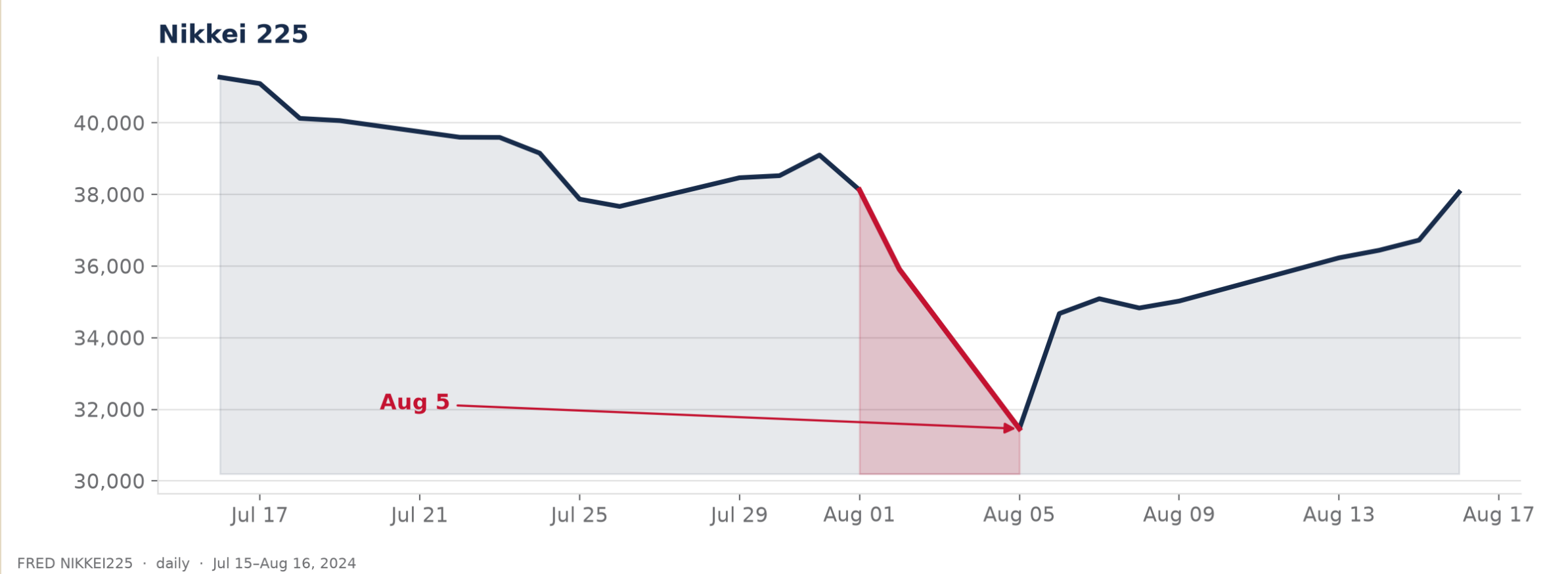
BLOCK 2

Why should a business student study macroeconomics?

Why does a minor adjustment in Tokyo matter to a portfolio manager in New York or an energy firm in Qatar?

Case study: the 2024 Nikkei mini-crash

Nikkei 225 tracks the Japanese stock market · July 31 – August 5, 2024



-17.2%

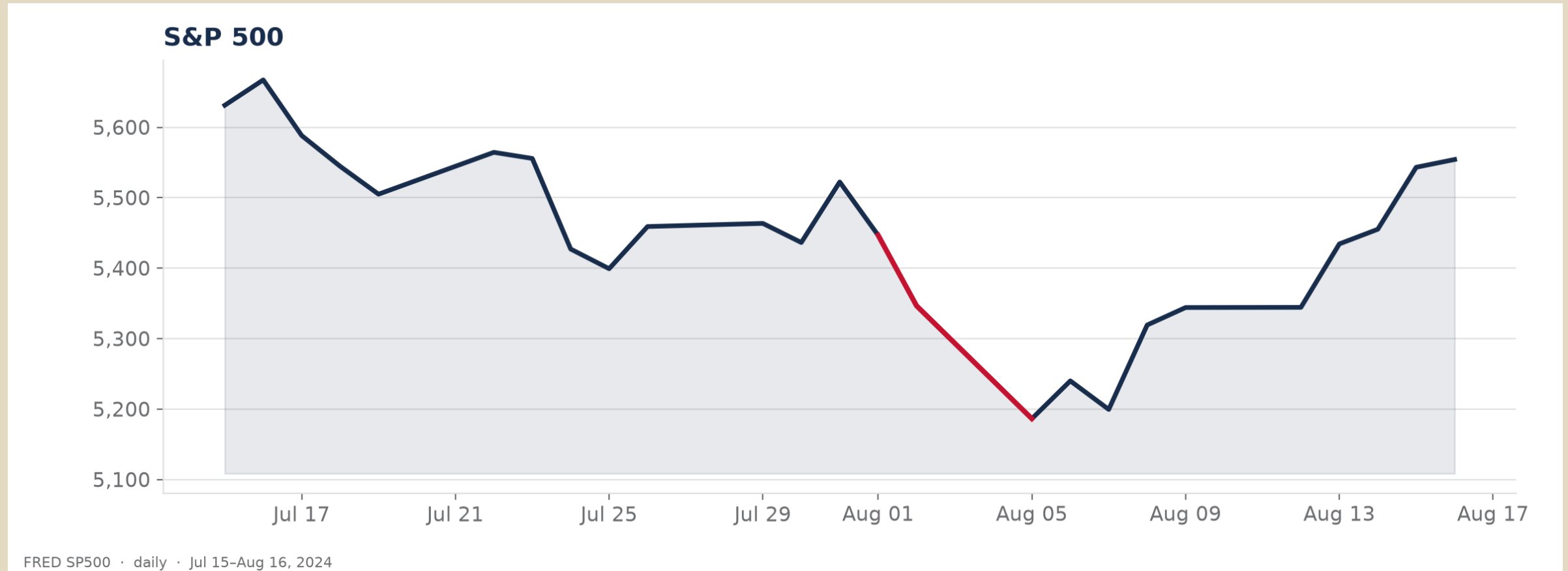
5-day drop. Sharpest decline in the Nikkei 225 since the 1987 Black Monday crash.

-33%

Decline during a potentially extinction-level event: COVID (Feb–March 2020).

Ripple effect into SPY

SPY tracks the S&P 500 · the same five days



-8% in 5 days

What happened?

The decision: a 0.15% increase in interest rates

THE DECISION

Bank of Japan's unexpected decision to raise its short-term policy interest rate.

Previous

0.1%



New rate

0.25%

A seemingly small 0.15% rate hike by the Bank of Japan triggered a massive global sell-off in stocks.

Global interconnectedness

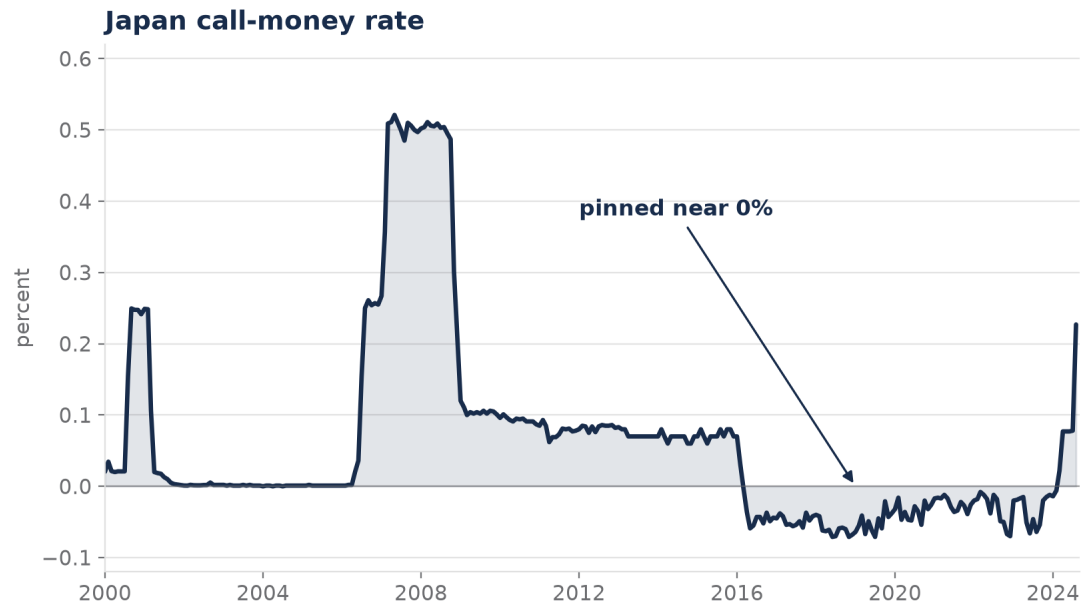
This event highlights how deeply integrated global financial markets have become.

The question

Why does a minor adjustment in Tokyo matter to a portfolio manager in New York or an energy firm in Qatar?

Before the how — some context

Japan has maintained low interest rates for some time



FRED IRSTCI01JPM156N · monthly · 2000-Jul 2024

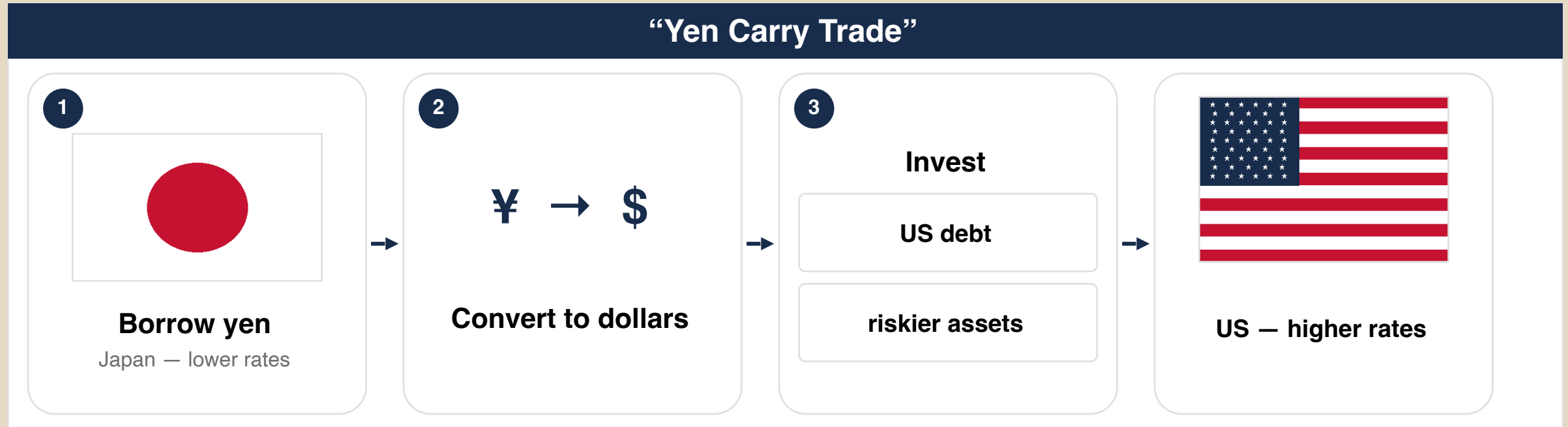
Yen losing value relative to the dollar



FRED DEXJPUS · daily · 2019-Aug 2024

Low interest rates lead to arbitrage opportunities

Low interest rates create incentives for investment. How?



**Also incentives for unnecessary
risk → leverage**

Remember, all this was happening while...

1. The yen was depreciating.
2. Inflation in Japan was getting out of control.

How to solve these issues?

Monetary policy!
→ ↑ interest rates

The Bank of Japan decided to increase interest rates

Why this could have caused the mini-crash?

1

Think about what makes the Yen Carry Trade work: stability in the exchange rate from yen to dollar.

2

What happened with the exchange rate once interest rates increased?

People over-leveraged will try to get out of the trade.

3

Margin calls

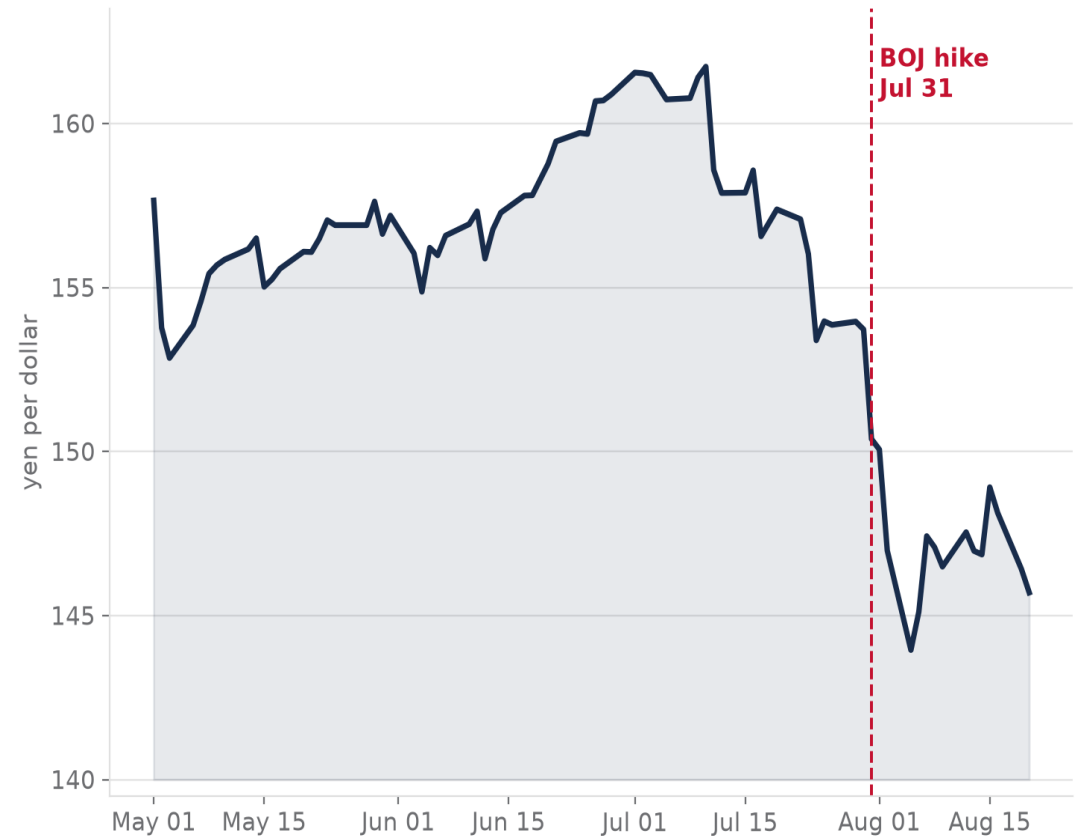


panic selling



mini crash

USD / JPY · the unwind



FRED DEXJPUS · daily · May-Aug 2024

Yen appreciated 12%

The carry-trade crisis — a step-by-step unwind

Case study: the August 2024 market volatility

01. Trade setup	02. Rate-hike shock	03. The crisis
¥100M Borrow at a low 0.1% interest rate.	0.25% BOJ raises rates on July 31, catching markets off-guard.	~¥90M You still owe ¥100M; USD assets now only buy ~¥90M.
USD 618K Convert at ¥162/\$.	12–14% Yen strengthens rapidly in just a few days.	¥10M Sudden loss triggers immediate margin calls.
~4.9% Invest in 5% U.S. assets — the carry.	¥146 USD/JPY drops from ¥162 to ¥146.	Crash Mass liquidation → global market crash.

What would you do?

SUPPLY CHAIN

What would you do as a supply chain manager for a Qatari energy firm during a global market crash like this?

MARKETING

What would you do as a marketing director for a Qatari tourism company amid economic uncertainty from such a crash?

Key takeaways

Why understanding these macroeconomic episodes should matter to you

1

Policy decisions

Macroeconomic policy decisions can immediately impact financial markets and other key economic variables.

2

Ripple effects

Effects quickly spread across sectors and economies, influencing critical business decisions on investing, marketing, and supply chains.

3

Global interconnectedness

Macroeconomics provides the framework to understand the complex interconnectedness of global markets and risks.

BLOCK 3

What does Macroeconomics study?

Objectives of Macroeconomic Policy

What are the objectives of macroeconomic policy?

01 Sustainable growth of GDP

02 High employment

03 Price stability

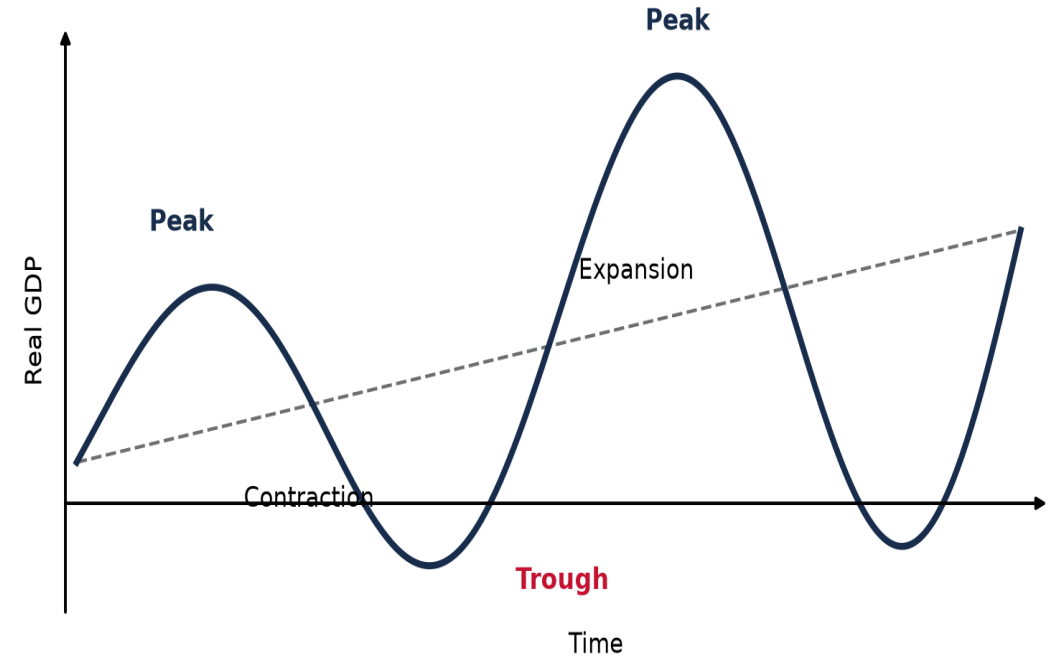
04 Exchange-rate stability

05 Income redistribution

So, what does Macroeconomics study?

- 1 Economic expansions and contractions
- 2 Production of goods and services
- 3 Unemployment
- 4 Price changes
- 5 Long-term economic growth and short-term fluctuations

Stages of the Business Cycle



Source: Federal Reserve Bank of St. Louis

A Map of the Semester

	Topic	Why this matters?	Tools
Measurement	GDP	This is the size of the economy. If you don't know what is in that number, you cannot follow the news or a forecast.	Circular flow · $Y \equiv C+I+G+NX$ · real vs nominal GDP · a statistical release
	Prices	Inflation changes what money can buy. Wages, prices, and contracts all depend on it.	CPI · GDP deflator · inflation rate · real vs nominal
	Labor	This is the job market: who is working, who is looking, how hard it is to hire.	$LF = E+U$ · unemployment rate · LFPR · employment–population ratio
Short-run demand	Goods / multiplier	In the short run, spending drives output. This is how a change in G or C becomes Y.	ZZ / goods-market equilibrium · multiplier · C, I, G, T · Blanchard 3
	Financial markets	Interest rates don't come from nowhere. This is how they're set, and why that matters for borrowing and investing.	Money demand $M_d = \bar{M} L(i)$ · $i = \bar{i}$ · open-market operations · Blanchard 4
	IS–LM	Spending and the interest rate move together. This is the map you use when policy or a shock hits both.	IS · LM (or $i = \bar{i}$) · fiscal/monetary mix · the dashboards
Medium run	Phillips	Jobs vs inflation is the argument you hear all the time. This is the relationship they mean.	Phillips curve · expected inflation · natural rate of unemployment · Blanchard 7–8
	Growth	Why some countries are rich and stay rich. That's a long-run question, not this quarter.	Production function $Y = F(K, N)$ · growth accounting · Handout 4 + R
Open	Open economy	We import, we export, we peg. The rest of the world is not optional here.	NX · real exchange rate · the peg · Blanchard 17–19

BLOCK 4

Structure of the Course and Expectations

Syllabus: Grades, AI, and expectations

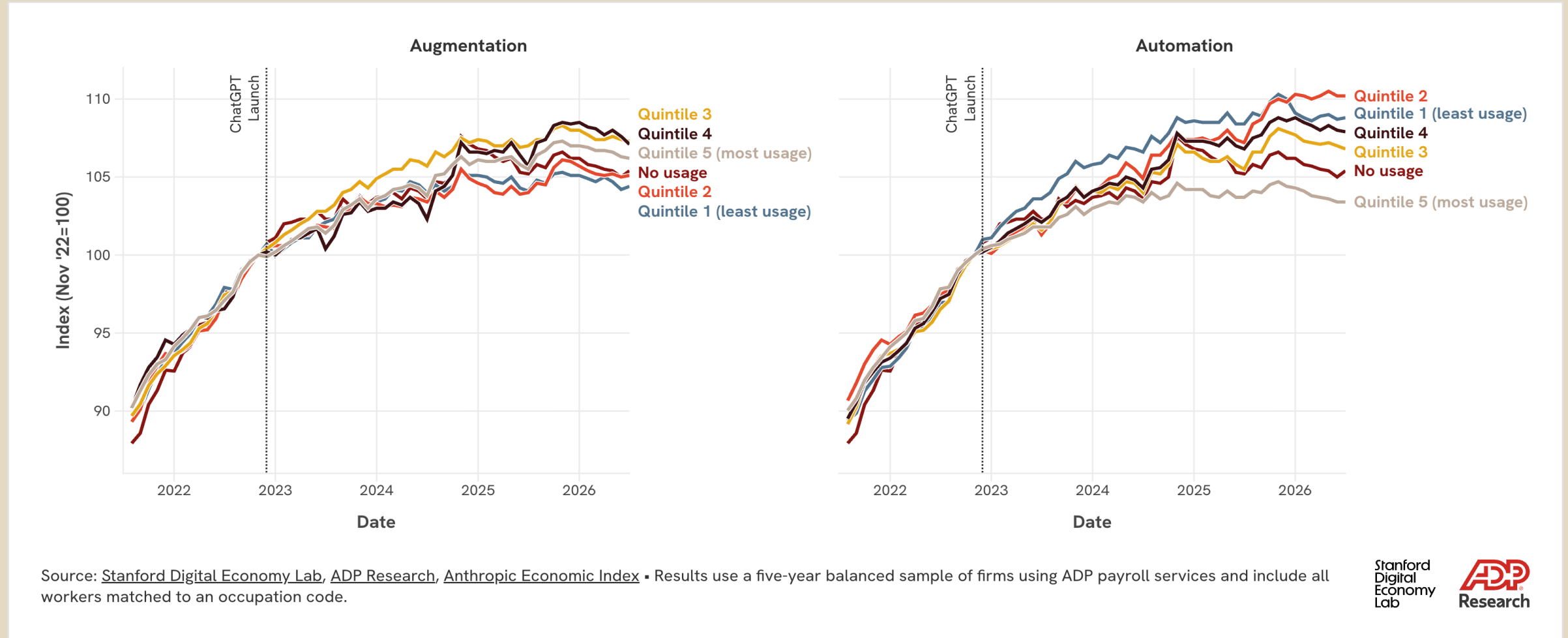
Course structure — grading breakdown

For full details please read the syllabus

Final exam	30%	During final examinations week
Midterm exam	25%	Wed 7 Oct, during class time
Quizzes	20%	6 quizzes total (lowest score dropped). Could take the form of in-class case studies. Tests preparedness for session.
Homework	15%	4 assignments total. 3 online submissions via Canvas. 1 class debate.
Attendance & participation	10%	Weekly article/video discussions. Posted on Canvas every Thursday. Deadline: Saturdays at 11:59pm.

AI tools are partners, not substitutes

Automation is cutting entry-level jobs; augmentation is not



Brynjolfsson, Chandar, and Chen, 2026 (rev.). Canaries in the Coal Mine. Stanford DEL / ADP. Not causal; US payroll panel.

AI tools are partners, not substitutes

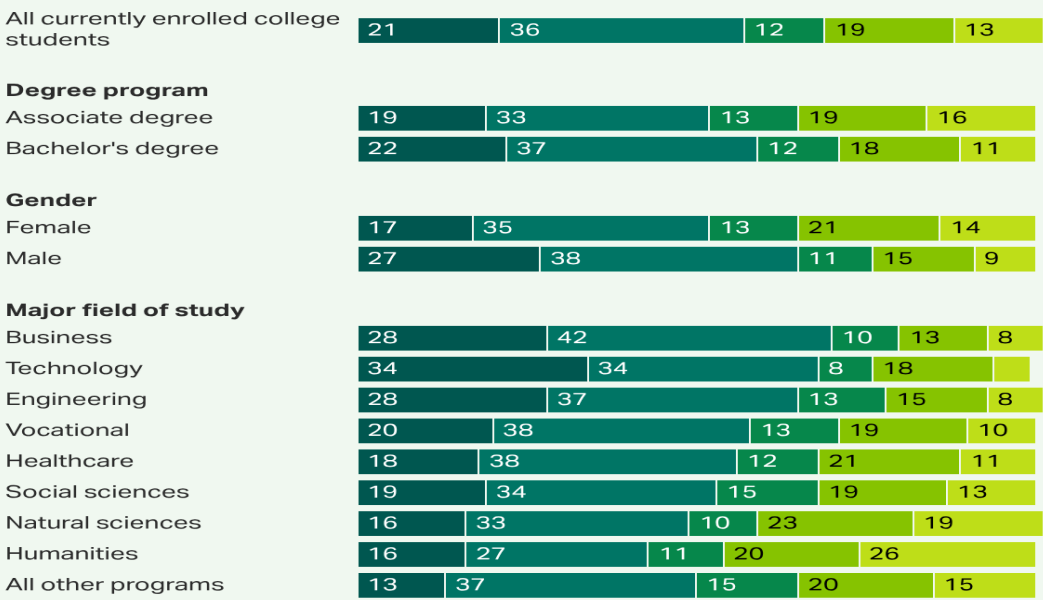
Most students use AI weekly: to learn, not just to finish faster

Majority of College Students Use AI in Coursework Weekly or Daily

How often do you use artificial intelligence (AI) in your coursework?

► Click to see how artificial intelligence was defined to respondents.

■ % Daily ■ % Weekly ■ % Monthly ■ % Infrequently ■ % Never



Oct. 2-31, 2025
Lumina Foundation-Gallup 2026 State of Higher Education Study

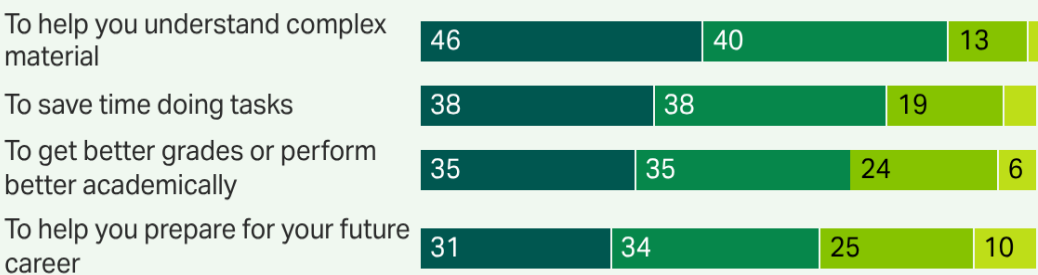
GALLUP

Students Cite Learning and Efficiency as Top Reasons for Using AI

How important are each of the following reasons why you [use/do not use] artificial intelligence in your coursework?

■ % Extremely important ■ % Very important ■ % Somewhat important ■ % Not important

Reasons for using AI



Among students who use AI for coursework on at least a monthly basis

Oct. 2-31, 2025
Lumina Foundation-Gallup 2026 State of Higher Education Study

GALLUP

Lumina Foundation-Gallup, 2026 State of Higher Education (fielded Oct 2-31, 2025). U.S. opt-in panel, not this room.

AI tools are partners, not substitutes

AI raises homework scores, and lowers exam scores

01

Homework up

Students who used AI scored higher on homework.

02

Time down

They finished faster — the work was outsourced.

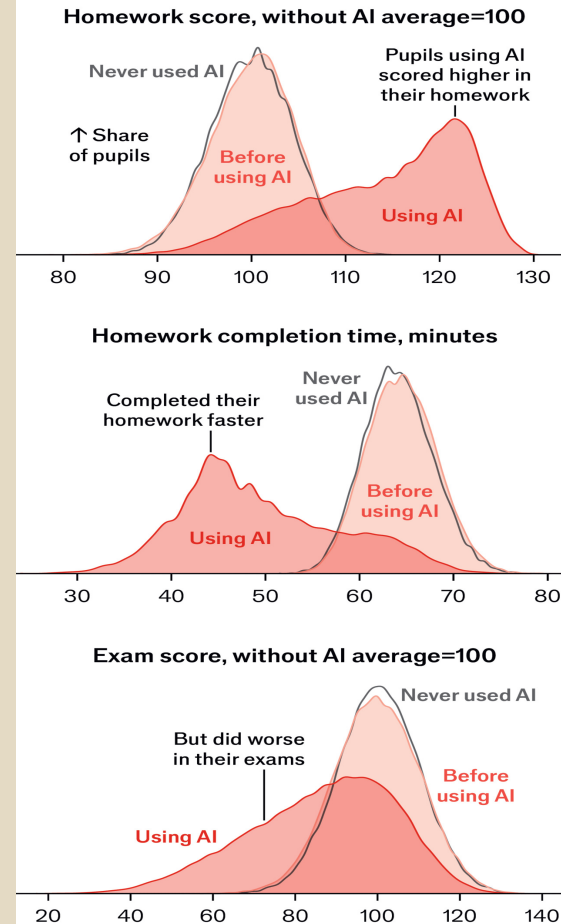
03

Exams down

On the exam, with the tool gone, they did worse.

AI ate my homework

China*, effect of using AI, Jan 2023-Jun 2025



*26,811 pupils aged 12-18

Source: "The generative AI learning penalty: evidence from Chinese secondary education", by D. Strömberg et al., preprint June 2026

AI and jobs: two predictions



Dario Amodei · CEO Anthropic

AI could eliminate half of entry-level white-collar jobs within 1–5 years... unemployment possibly 10–20%.

Cognitive substitute.

Speed + breadth make adaptation unusually painful.

Axios, 28 May 2025. “The Adolescence of Technology,” Jan 2026.



Jensen Huang · CEO NVIDIA

You’re not going to lose your job to an AI, but you’re going to lose your job to someone who uses AI.

Tasks are automated; the purpose of the job expands.

Productivity multiplies demand → more jobs and new occupations.

Milken Institute, 6 May 2025. CNBC Make It, 28 May 2025.

[Video](#)

Which prediction do you hope is true? Which one do you expect?

What does each view ask of you as future analysts and managers?

Main takeaways

01

Most organizations are using AI.

Very few can show it added value.

02

Firms are hiring people who can work with AI,

not people AI can replace.

03

The tool can help you learn.

It cannot replace understanding, checking the work, or judgment.

You're not going to lose your job to an AI. You're going to lose it to someone who uses AI to become more curious, to understand ideas and methods more deeply, to exercise better judgment, and to communicate more effectively.

Adapted from Jensen Huang, May 2025 · the first two sentences are his; the rest is what “uses” means in 73-103.

Effort is correlated with performance, but...

The New York Times

Opinion · Guest Essay · Adam Grant · 26 Dec 2024

No, You Don't Get an A for Effort

[Read the full article on NYTimes.com](#)

The Path to Average



Effort alone MIGHT help you get here.

Showing up and submitting can get you onto this path. They are not the same as command of the arguments.

The Path to Excellence



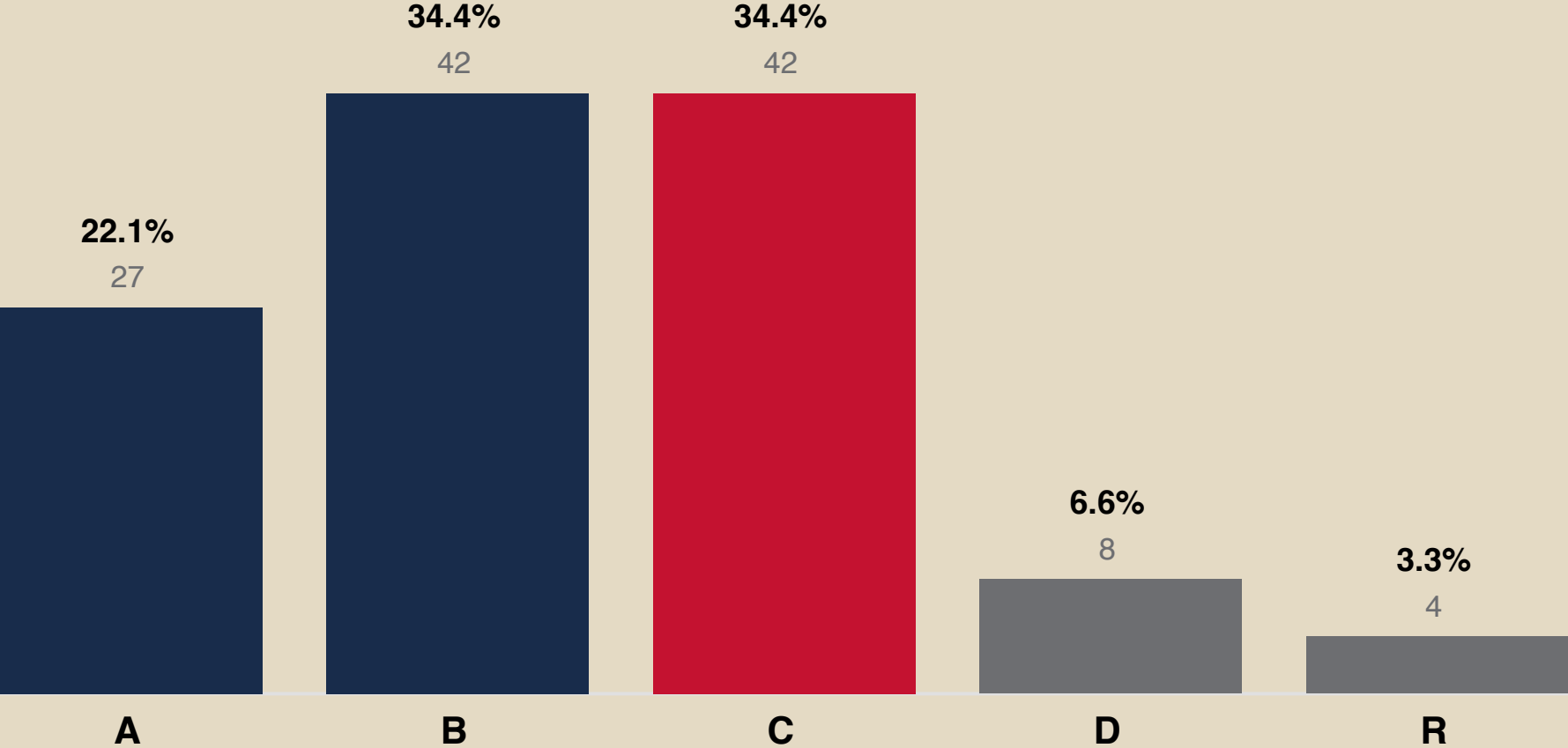
Effort

+ Diligence (studying, attending, submitting)

+ Mastery of the material

Grade distribution for the course

6 cohorts, n = 123



Source: instructor records, 6 cohorts, n = 122.

Recommendations

Strategies to perform well in this course

01

Understand grading criteria

Familiarize yourself with the syllabus and rubric. Comprehend exactly what is expected for each assignment and exam.

02

Master the material

Effort alone is not enough. While effort is valued, achieving top grades requires a deeper mastery of the subject matter.

03

Seek feedback and act

Regularly consult with me or the TA. Use constructive feedback to identify areas for improvement and adjust your approach.

04

Attend and engage

Come ready to engage. Actively participate in class discussions and group activities. Presence matters.

05

Use available resources

Don't struggle alone. Come to office hours (professor, TA, or CA) and ask questions whenever you are unsure.

KEY MESSAGE

This class will challenge you to think critically about macroeconomic problems...

...but rest assured, we're here to support you every step of the way.

Resources available to CMU-Q students

On campus / news

CMU-Q Library

Bloomberg

The Wall Street Journal

Rocketblocks.me

Management Consulted

Sources of data

American Economic Association data sources (aggregated)

IPUMS (census and survey data)

FRED — Federal Reserve Bank of St. Louis (economic time series)

Our World in Data

World Bank data (socioeconomic indicators for many countries)

IMF data (macro-focused)

Qatar Central Bank

Looking ahead

NEXT CLASS

GDP · CORE 3.1–3.4. We start measuring what the economy produced.

HOMEWORK

HW1 out Wed 26 Aug.

Answers to discussion questions for two macroeconomic articles shared through Canvas.

References

Sources used on Monday's slides. You can look these up.

Case study and data

Federal Reserve Bank of St. Louis, FRED. Nikkei and S&P 500 series used in the yen-carry case.

[Foreign Policy, 8 August 2024. William Sposato, “How Japan’s Yen Carry Trade Crashed Global Markets.” https://foreignpolicy.com/2024/08/08/japan-crash-yen-carry-trade-global-markets/](https://foreignpolicy.com/2024/08/08/japan-crash-yen-carry-trade-global-markets/)

News exhibits on the macroeconomic-problems slide: The Wall Street Journal (inflation / food and gasoline); Bloomberg (UK public debt); QNB (global trade). As shown in class.

Federal Reserve Bank of St. Louis also appears as the source on the “what does macroeconomics study” figure.

References

AI, jobs, and effort — same claims as in class.

AI, jobs, and effort

Brynjolfsson, Chandar, and Chen, 2026 (rev.). Canaries in the Coal Mine. Stanford Digital Economy Lab / ADP Research. Not causal; US payroll panel.

Lumina Foundation–Gallup, 2026 State of Higher Education (fielded 2–31 October 2025). U.S. opt-in panel, not this classroom.

Strömberg, Lei, and Wu, 2026. CEPR Discussion Paper 21577. China, grades 7–12, n = 26,811. Not this classroom.

[Dario Amodei on entry-level white-collar jobs: Axios, 28 May 2025. <https://www.axios.com/2025/05/28/ai-jobs-white-collar-unemployment-anthropic>. Also “The Adolescence of Technology,” January 2026.](https://www.axios.com/2025/05/28/ai-jobs-white-collar-unemployment-anthropic)

[Jensen Huang: Milken Institute, 6 May 2025; CNBC Make It, 28 May 2025. Video: <https://www.youtube.com/watch?v=HT8-KPAjpiA>](https://www.youtube.com/watch?v=HT8-KPAjpiA)

[Adam Grant, New York Times guest essay, 26 December 2024. <https://www.nytimes.com/2024/12/26/opinion/school-grades-a-quantity-quality.html>](https://www.nytimes.com/2024/12/26/opinion/school-grades-a-quantity-quality.html)